IEEE BAU COMPUTER SOCIETY

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# CIPHER

## Network detection and response

A security prototype for small offices, shops, and schools. CIPHER learns each device's normal traffic pattern, flags unusual behavior, and explains the evidence so an operator can choose a response.

### FROM NETWORK TRAFFIC TO OPERATOR ACTION

**01 OBSERVE**
**Linux router**
eBPF / XDP
Traffic hook

→

**02 SUMMARIZE**
**Rust sensor**
Sampled traffic
metadata

→

**03 ANALYZE**
**Rust server**
Statistics +
AI review

→

**04 RESPOND**
**Web dashboard**
Review +
device control

FlatBuffers reports over a TLS WebSocket · Axum server · SQLite persistence

## How detection works

**50 reports per device** establish a baseline of normal activity.

**Four signals:** incoming bytes, outgoing bytes, dropped events, and port entropy (how varied port use is).

**Statistics flag the change.** An optional AI reviewer explains the evidence and recommends a response.

Measurements are sampled observations. An anomaly is a reason to investigate, not proof of an attack.

## Operator control

**Monitor and alert is the default.** Operators can trust devices, review incidents, and manually isolate or restore a device.

**Cloud auto-isolation** requires enforcement mode, an untrusted device, at least two abnormal signals, and a valid block recommendation scored at least 85/100. The incident must remain open and policy unchanged.

The score is the model's own assessment, not a measured probability.

Local edge rules are separate. Monitor mode does not disable packet filters or clear existing blocks.

### PROTOTYPE AND DEMO

**Working software:** detection, incident review, saved baselines, trust controls, acknowledgments, and retries.

**Booth demo:** simulated sensor and scripted reviewer with the real backend; no live packet blocking. Hardware enforcement and detection accuracy still need validation.

**See the demo and explore the project**

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The Four Horsemen · @ieee_bau_cs

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